

C363

# Calculus I

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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## How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A physics teacher who also coaches the robotics club asks her class to find the limit as  $x$  approaches 3 of  $(x^2 - 9)/(x - 3)$ . The class uses graphing calculators donated by a local bank. What is the limit?

- A. 6
- B. 3
- C. 0
- D. The limit does not exist because the denominator is 0

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

An engineer at a firm with 300 employees models a load with  $f(x) = x^2 + 1$  for  $x < 2$  and  $f(x) = kx - 3$  for  $x \geq 2$ . The design review is scheduled for Friday. What value of  $k$  makes  $f$  continuous at  $x = 2$ ?

- A. Any  $k$ , because  $f(2)$  is defined
- B.  $k = 1$
- C.  $k = 5$
- D.  $k = 4$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

A data analyst who joined the team last month needs the derivative of  $f(x) = (3x^2 + 1)^5$  for a sensitivity report due to a client in Denver. What is  $f'(x)$ ?

- A.  $30x(3x^2 + 1)^5$
- B.  $30x(3x^2 + 1)^4$
- C.  $5(3x^2 + 1)^4$
- D.  $5(6x)^4$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4. [TRANSFER — reversed stem / different variable]

Let  $h(x) = f(g(x))$  with  $g(1) = 2$ ,  $g'(1) = 3$ ,  $f(1) = 7$ , and  $f'(2) = 4$ . What is  $h'(1)$ ?

- A. 7
- B. 4
- C. 12
- D. 21

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5.

A small online retailer with a loyal customer base models average revenue as  $f(x) = x^2 / (x + 1)$ , where  $x$  is in thousands of visits. Its founder studied economics in college. What is  $f'(1)$ ?

- A.  $3/4$
- B. 2
- C.  $1/2$
- D.  $-3/4$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

A particle's position in meters is  $s(t) = t^3 - 6t^2 + 9t$ . The lab running the experiment is funded by a state grant and uses sensors accurate to 0.01 m. At  $t = 0.5$  seconds, what is the particle doing?

- A. Moving in the negative direction
- B. At rest
- C. Speeding up, because its velocity is positive
- D. Slowing down

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7. [TRANSFER — reversed stem / different variable]

A particle has velocity  $v(t) = t^2 - 4$  for  $t \geq 0$ . At what time is it at rest?

- A.  $t = 0$
- B.  $t = 2$
- C.  $t = 2$  and  $t = -2$
- D.  $t = 4$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

A farmer who has worked the same land for 25 years has 400 meters of fencing to enclose a rectangular pasture along a straight river; no fence is needed on the river side. Her neighbor used wooden posts. What is the largest possible area?

- A. 10,000 m<sup>2</sup>
- B. 20,000 m<sup>2</sup>
- C. 100 m
- D. About 17,778 m<sup>2</sup>

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

9. [TRANSFER — reversed stem / different variable]

On the closed interval  $[0, 5]$ ,  $f$  has one critical point at  $x = 2$  with  $f(2) = 3$ . Also  $f(0) = 1$  and  $f(5) = 8$ . What is the absolute maximum value?

- A. 5
- B. There is none, since the critical point is not a maximum
- C. 3
- D. 8

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

10.

A pebble dropped in a pond at a city park that was renovated in 2019 creates a circular ripple whose radius grows at 2 cm per second. The water is about 1.5 meters deep. How fast is the area changing when the radius is 5 cm?

- A.  $4\pi$  cm<sup>2</sup>/s
- B.  $10\pi$  cm<sup>2</sup>/s
- C.  $20\pi$  cm<sup>2</sup>/s
- D.  $25\pi$  cm<sup>2</sup>/s

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

11.

A teacher preparing a review for 30 students writes the limit as  $x$  approaches 0 of  $\sin(3x)/x$ . She has taught calculus for eight years. What is the limit?

- A. Undefined, because you cannot divide by zero
- B. 0
- C. 3
- D. 1

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

12. [TRANSFER — reversed stem / different variable]

To which limit can L'Hôpital's rule NOT be applied directly?

- A. The limit as  $x \rightarrow 1$  of  $(x^2 + 1)/(x + 1)$
- B. The limit as  $x \rightarrow \infty$  of  $x/e^x$
- C. The limit as  $x \rightarrow 0$  of  $(e^x - 1)/x$
- D. The limit as  $x \rightarrow 2$  of  $(x^2 - 4)/(x - 2)$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer A Concept 1

A physics teacher ~~who also coaches the robotics club~~ asks her class to find the limit as  $x$  approaches 3 of  $(x^2 - 9)/(x - 3)$ . ~~The class uses graphing calculators donated by a local bank.~~ What is the limit?

**Target word:** value of the limit **Decides:** Factor:  $(x - 3)(x + 3)/(x - 3) = x + 3$ .

|   |                   |                                                                                       |
|---|-------------------|---------------------------------------------------------------------------------------|
| A | <b>Correct</b>    | After canceling, the limit is $3 + 3 = 6$ .                                           |
| B | Wrong stem number | Reports the $x$ -value being approached.                                              |
| C | Dropped element   | Substitutes only into the numerator.                                                  |
| D | Absolute language | A $0/0$ form can still have a limit; the function has a hole, not a gap in the limit. |

## 2. Answer D Concept 2

An engineer ~~at a firm with 300 employees~~ models a load with  $f(x) = x^2 + 1$  for  $x < 2$  and  $f(x) = kx - 3$  for  $x \geq 2$ . ~~The design review is scheduled for Friday.~~ What value of  $k$  makes  $f$  continuous at  $x = 2$ ?

**Target word:**  $k$  for continuity **Decides:** The pieces must meet:  $2^2 + 1 = 2k - 3$ .

|   |                    |                                                        |
|---|--------------------|--------------------------------------------------------|
| A | Dropped element    | Drops the condition that the limit must equal $f(2)$ . |
| B | Inverted operation | Subtracts 3 instead of adding it: $2k = 2$ .           |
| C | Wrong stem number  | Uses the left-hand limit 5 as $k$ .                    |
| D | <b>Correct</b>     | $5 = 2k - 3$ gives $2k = 8$ , so $k = 4$ .             |

## 3. Answer B Concept 3

A data analyst ~~who joined the team last month~~ needs the derivative of  $f(x) = (3x^2 + 1)^5$  for a sensitivity report ~~due to a client in Denver.~~ What is  $f'(x)$ ?

**Target word:** derivative of a composite **Decides:** Chain rule: multiply by the inner derivative  $6x$ .

|   |                   |                                                                  |
|---|-------------------|------------------------------------------------------------------|
| A | Wrong stem number | Keeps the original exponent instead of lowering it by one.       |
| B | <b>Correct</b>    | $5(3x^2 + 1)^4 \cdot 6x = 30x(3x^2 + 1)^4$ .                     |
| C | Dropped element   | Drops the inner derivative $6x$ .                                |
| D | Confusion set     | Differentiates the inside and then applies the power rule to it. |

## 4. Answer C Concept 3 • transfer

Let  $h(x) = f(g(x))$  with  $g(1) = 2$ ,  $g'(1) = 3$ ,  $f(1) = 7$ , and  $f'(2) = 4$ . What is  $h'(1)$ ?

**Target word:** chain rule from a table **Decides:**  $h'(1) = f'(g(1)) \cdot g'(1) = f'(2) \cdot 3$ .

|   |                   |                                                                 |
|---|-------------------|-----------------------------------------------------------------|
| A | Wrong stem number | Reports $f'(1)$ , a number that does not enter the computation. |
| B | Dropped element   | Drops the factor $g'(1)$ .                                      |
| C | <b>Correct</b>    | $f'(2) \cdot g'(1) = 4 \cdot 3 = 12$ .                          |
| D | Confusion set     | Uses $f'(1)$ instead of $f'(g(1)) = f'(2)$ .                    |



## 5. Answer A Concept 4

A small online retailer with a loyal customer base models average revenue as  $f(x) = x^2 / (x + 1)$ , where  $x$  is in thousands of visits. ~~Its founder studied economics in college.~~ What is  $f'(1)$ ?

**Target word:** derivative value at  $x = 1$  **Decides:** Quotient rule:  $(2x(x + 1) - x^2)/(x + 1)^2$ .

|   |                    |                                                                                     |
|---|--------------------|-------------------------------------------------------------------------------------|
| A | <b>Correct</b>     | $(2 \cdot 1 \cdot 2 - 1)/2^2 = 3/4$ .                                               |
| B | Confusion set      | Divides the derivative of the top ( $2x$ ) by the derivative of the bottom ( $1$ ). |
| C | Adjacent category  | That is $f(1)$ , the function value, not the derivative.                            |
| D | Inverted operation | Reverses the numerator order: $uv' - u'v$ .                                         |

## 6. Answer D Concept 5

A particle's position in meters is  $s(t) = t^3 - 6t^2 + 9t$ . The lab running the experiment ~~is funded by a state grant and uses sensors accurate to 0.01 m.~~ At  $t = 0.5$  seconds, what is the particle doing?

**Target word:** speeding up or slowing down **Decides:**  $v(0.5) = 3.75 > 0$  and  $a(0.5) = -9 < 0$ : opposite signs.

|   |                    |                                                                                   |
|---|--------------------|-----------------------------------------------------------------------------------|
| A | Inverted operation | Velocity is positive, so it moves in the positive direction.                      |
| B | Confusion set      | At rest requires $v = 0$ ; $v = 3.75$ here.                                       |
| C | Dropped element    | Ignores acceleration; speed depends on both signs.                                |
| D | <b>Correct</b>     | $v = 3t^2 - 12t + 9 = 3.75$ and $a = 6t - 12 = -9$ ; opposite signs mean slowing. |

## 7. Answer B Concept 5 · transfer

A particle has velocity  $v(t) = t^2 - 4$  for  $t \geq 0$ . At what time is it at rest?

**Target word:** time when  $v = 0$  **Decides:**  $t^2 - 4 = 0$  with  $t \geq 0$ .

|   |                   |                                                |
|---|-------------------|------------------------------------------------|
| A | Confusion set     | That is the start, when $v = -4$ , not rest.   |
| B | <b>Correct</b>    | $t^2 = 4$ and $t \geq 0$ leaves only $t = 2$ . |
| C | Adjacent category | Negative time is outside the given domain.     |
| D | Wrong stem number | Uses the constant 4 as the time.               |

## 8. Answer B Concept 6

A farmer ~~who has worked the same land for 25 years~~ has 400 meters of fencing to enclose a rectangular pasture along a straight river; no fence is needed on the river side. ~~Her neighbor used wooden posts.~~ What is the largest possible area?

**Target word:** maximum area **Decides:**  $2x + y = 400$ ,  $A = x(400 - 2x)$ , max at  $x = 100$ .

|   |                   |                                                                             |
|---|-------------------|-----------------------------------------------------------------------------|
| A | Dropped element   | Fences all four sides (a $100 \times 100$ square), dropping the river side. |
| B | <b>Correct</b>    | $x = 100$ , $y = 200$ ; area = $100 \times 200 = 20,000$ .                  |
| C | Confusion set     | The optimal width, not the area.                                            |
| D | Adjacent category | Assumes a square ( $400/3$ per side) is always best.                        |

### 9. Answer D Concept 6 · transfer

On the closed interval  $[0, 5]$ ,  $f$  has one critical point at  $x = 2$  with  $f(2) = 3$ . Also  $f(0) = 1$  and  $f(5) = 8$ . What is the absolute maximum value?

**Target word:** absolute max on a closed interval **Decides:** Compare the critical value with both endpoints.

|   |                   |                                                                            |
|---|-------------------|----------------------------------------------------------------------------|
| A | Confusion set     | Reports the $x$ -location of the maximum, not the value.                   |
| B | Absolute language | A continuous function on a closed interval always has an absolute maximum. |
| C | Dropped element   | Skips the endpoints.                                                       |
| D | <b>Correct</b>    | Candidates are 1, 3, and 8; the largest is 8, at $x = 5$ .                 |

### 10. Answer C Concept 7

A pebble dropped in a pond at a city park ~~that was renovated in 2019~~ creates a circular ripple whose radius grows at 2 cm per second. ~~The water is about 1.5 meters deep.~~ How fast is the area changing when the radius is 5 cm?

**Target word:** rate of area change **Decides:**  $dA/dt = 2\pi r \cdot dr/dt$ .

|   |                   |                                                       |
|---|-------------------|-------------------------------------------------------|
| A | Wrong stem number | Substitutes the rate 2 for the radius.                |
| B | Dropped element   | Drops the factor 2 from the derivative of $r^2$ .     |
| C | <b>Correct</b>    | $2\pi(5)(2) = 20\pi$ .                                |
| D | Adjacent category | That is the area at $r = 5$ , not its rate of change. |

### 11. Answer C Concept 8

A teacher preparing a review ~~for 30 students~~ writes the limit as  $x$  approaches 0 of  $\sin(3x)/x$ . ~~She has taught calculus for eight years.~~ What is the limit?

**Target word:** value of a  $0/0$  limit **Decides:**  $0/0$  form; L'Hôpital:  $3\cos(3x)/1 \rightarrow 3$ .

|   |                   |                                                          |
|---|-------------------|----------------------------------------------------------|
| A | Absolute language | $0/0$ is indeterminate, not undefined; the limit exists. |
| B | Dropped element   | Substitutes only into the numerator.                     |
| C | <b>Correct</b>    | Differentiate top and bottom: $3\cos(0)/1 = 3$ .         |
| D | Confusion set     | Recalls $\sin(x)/x \rightarrow 1$ and ignores the 3.     |

### 12. Answer A Concept 8 · transfer

To which limit can L'Hôpital's rule NOT be applied directly?

**Target word:** reverse: find the non-indeterminate form **Decides:** Only one limit is not  $0/0$  or  $\infty/\infty$ .

|   |                    |                                                                 |
|---|--------------------|-----------------------------------------------------------------|
| A | <b>Correct</b>     | Substitution gives $2/2 = 1$ ; it is not an indeterminate form. |
| B | Adjacent category  | $\infty/\infty$ also qualifies for the rule.                    |
| C | Confusion set      | This is $0/0$ , so the rule applies.                            |
| D | Inverted operation | $0/0$ ; factoring works, but the rule also applies.             |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 9             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 6             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 5             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 4             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 3             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 9             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 0             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                    | Rule                                                                                                                                          | Trap                                                                                       | Tell                                                                  |
|---|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1 | <b>Limits at removable discontinuities</b> | If direct substitution gives $0/0$ , factor and cancel (or simplify) before substituting; the limit can exist where the function does not.    | Calling any $0/0$ result zero or undefined.                                                | A rational expression whose numerator and denominator share a factor. |
| 2 | <b>Continuity at a point</b>               | $f$ is continuous at $a$ if $f(a)$ exists, the limit exists, and they are equal; for a piecewise function the pieces must meet.               | Thinking a defined $f(a)$ alone makes $f$ continuous.                                      | "Find $k$ so that $f$ is continuous."                                 |
| 3 | <b>Chain rule</b>                          | $d/dx f(g(x)) = f'(g(x)) \cdot g'(x)$ : differentiate the outside, keep the inside, multiply by the inside's derivative.                      | Forgetting to multiply by the inner derivative.                                            | A function raised to a power or nested inside another.                |
| 4 | <b>Product and quotient rules</b>          | $(u/v)' = (u'v - uv')/v^2$ and $(uv)' = u'v + uv'$ ; the derivative of a quotient is not the quotient of derivatives.                         | Dividing $u'$ by $v'$ , or reversing the order of the quotient-rule numerator.             | A fraction with $x$ in both top and bottom.                           |
| 5 | <b>Position, velocity, acceleration</b>    | $v = s'$ and $a = v'$ ; an object speeds up when $v$ and $a$ have the same sign and slows down when they differ; it is at rest when $v = 0$ . | Judging speeding up from velocity alone.                                                   | "Speeding up or slowing down?" or "at rest".                          |
| 6 | <b>Optimization and extreme values</b>     | Absolute extremes on a closed interval occur at critical points or endpoints; set up the objective, use the constraint, then solve $f' = 0$ . | Checking only critical points, or reporting the $x$ -value instead of the extreme value.   | "Maximize" or "minimize" with a fixed resource.                       |
| 7 | <b>Related rates</b>                       | Write an equation linking the quantities, differentiate both sides with respect to time, then substitute the instant's values.                | Substituting values before differentiating, or reporting the quantity instead of its rate. | "How fast is ... changing when ...?"                                  |
| 8 | <b>L'Hôpital's rule</b>                    | L'Hôpital's rule applies only to $0/0$ or $\infty/\infty$ forms: then the limit equals the limit of $f'/g'$ .                                 | Using it on any fraction, or applying the quotient rule instead.                           | A limit that substitutes to $0/0$ or $\infty/\infty$ .                |

**Course scope (public description):** Calculus I is the study of rates of change in the slope of a curve and covers the knowledge and skills necessary to apply differential calculus of one variable and to use appropriate technology to model and solve real-life problems. Topics include functions, limits, continuity, differentiability, visual, analytical, and conceptual approaches to the definition of the derivative; the power, chain, sum, product, and quotient rules applied to polynomial, trigonometric, exponential, and logarithmic functions; implicit differentiation, position, velocity, and acceleration; optimization, related rates, curve sketching, and L'Hopital's rule. Precalculus is a prerequisite for this course.